



School of Information Technology

Operating Systems and Administration

Practical 5 – Windows Security Analysis and Configuration Assignment (30%)

2025 / 2026 Semester 2

APPENDIX A - TEAM ORGANISATION

Module Group : IT2164

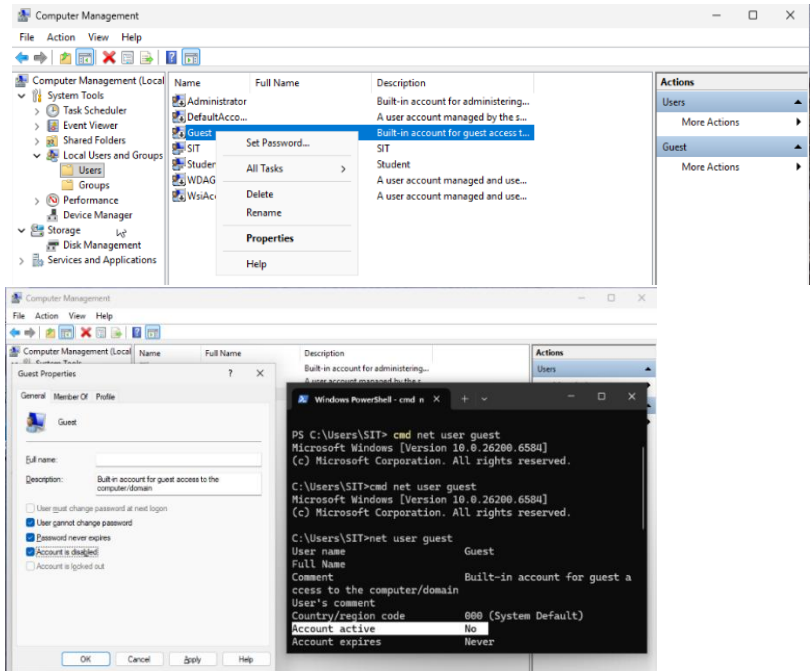
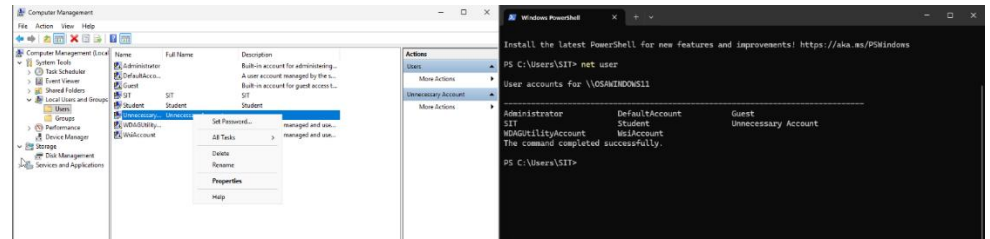
Team No : 01

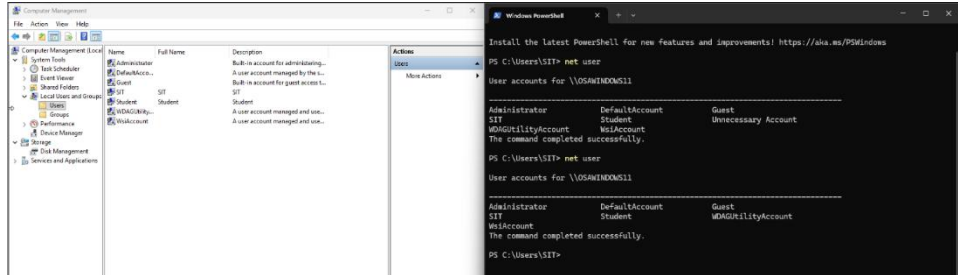
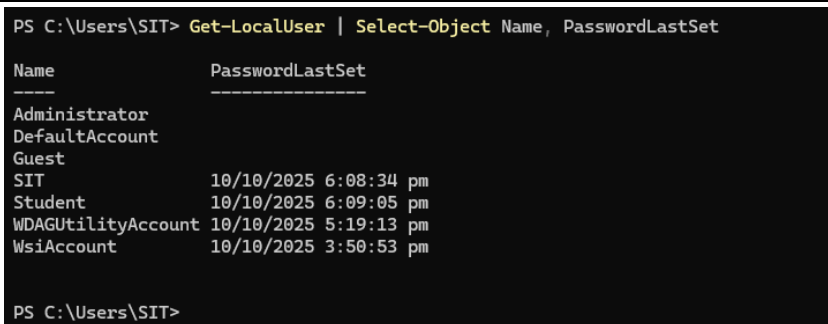
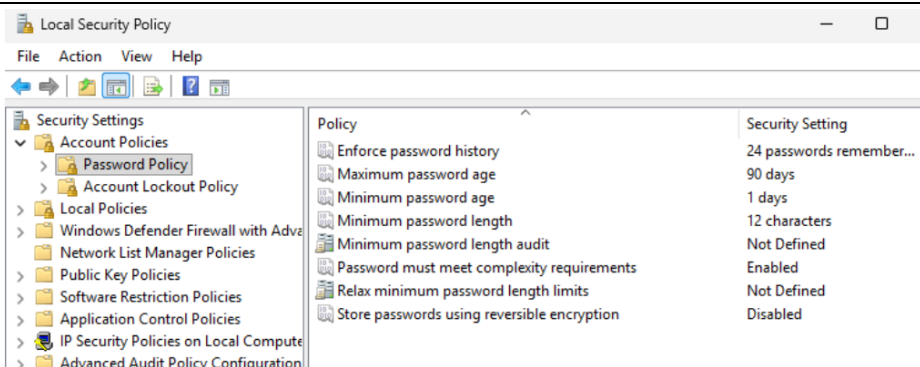
Scenario : B

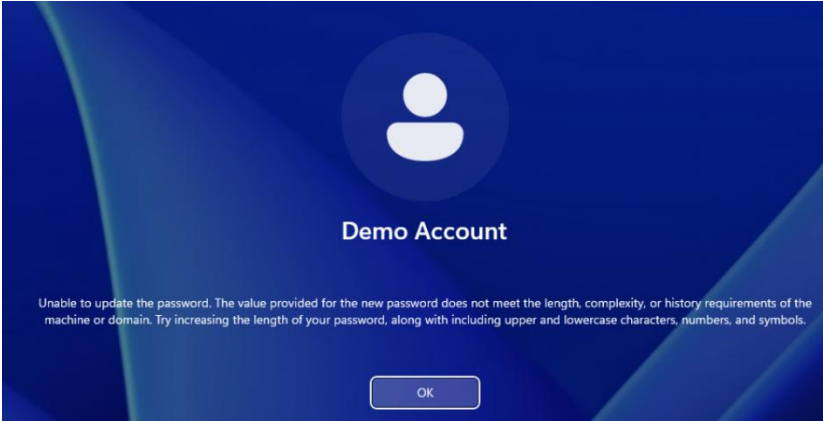
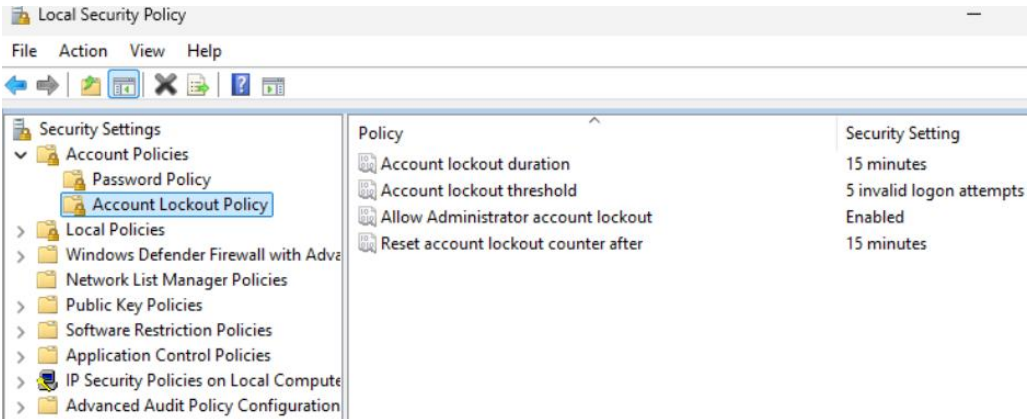
Team leader and members:

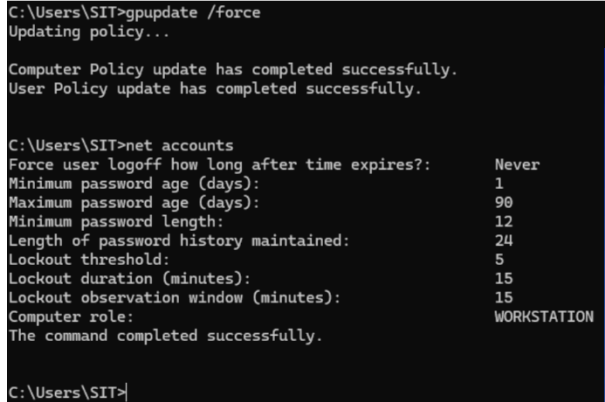
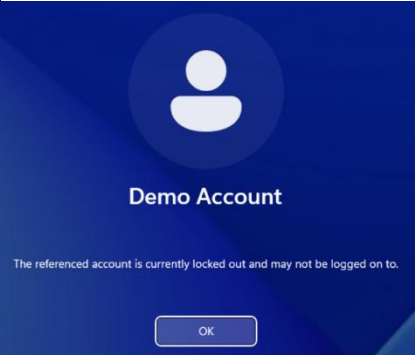
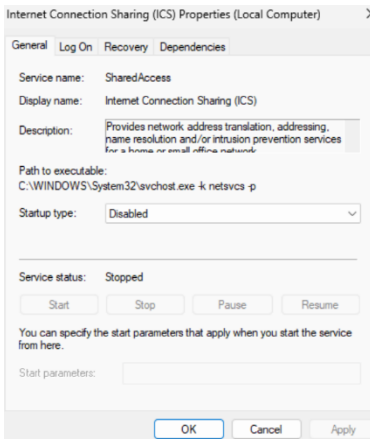
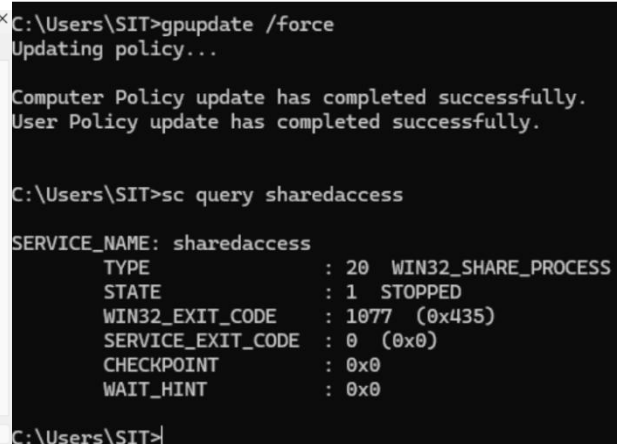
S/No	Admin No.	Name	Role
1.	240338L	Li Jingda	Account Security Lead
2.	212059D	Ong Yong Lin	System Policy Lead
3.	231150Z	Darren Tham	Network & Services Lead
4.	243402M	Zwe Htet Kyaw	Software Update & Security Lead

APPENDIX B (Part 1) - Baseline Security Checklists

	Windows Baseline Security Issues	Done By	Action/Solution (short description and/or screenshot)
1	Make sure the Guest account is disabled	Jingda	 Prevent unauthorised access with no account Computer management > Local Users and Groups > Users > Guests > Properties Check Account is Disabled, net user guest to verify Guest account is no longer active
2	Disable or delete any unnecessary accounts	Jingda	

			 Computer management > Local Users and Groups > Users > Guests > Delete To check: net user (returns a list of users in the OS)
3	Accounts are protected with passwords	Jingda	 Ensures all active accounts require secure authentication. Get-LocalUser Select-Object Name, PasswordLastSet Computer management > Local Users and Groups > Users > /Account Name/ > Set Password
4	Set strong password policies	Yong Lin	

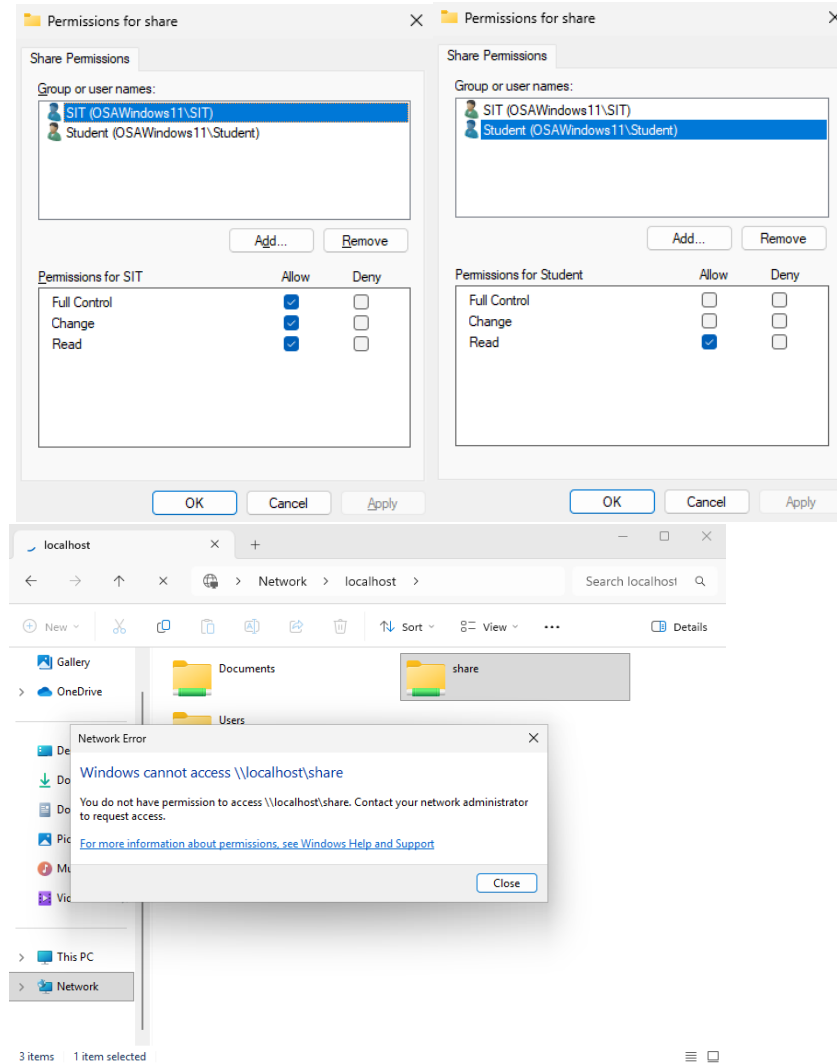
			<pre> C:\Users\SIT>gpupdate /force Updating policy... Computer Policy update has completed successfully. User Policy update has completed successfully. C:\Users\SIT>net accounts Force user logoff how long after time expires?: Never Minimum password age (days): 1 Maximum password age (days): 90 Minimum password length: 12 Length of password history maintained: 24 Lockout threshold: 10 Lockout duration (minutes): 10 Lockout observation window (minutes): 10 Computer role: WORKSTATION The command completed successfully. C:\Users\SIT> </pre>  I have configured the following: • Minimum password length = 12 • Password history = 24 passwords (prevent reuse of recent passwords) • Minimum password age = 1 day • Maximum password age = 90 days (Forces periodic password changes) • Set “Password must meet complexity requirement” to Enabled (This means all password must contain uppercase, lowercase, numbers and symbols)
5	Set account lockout policy	Yong Lin	

			 <pre> C:\Users\SIT>gpupdate /force Updating policy... Computer Policy update has completed successfully. User Policy update has completed successfully. C:\Users\SIT>net accounts Force user logoff how long after time expires?: Never Minimum password age (days): 1 Maximum password age (days): 90 Minimum password length: 12 Length of password history maintained: 24 Lockout threshold: 5 Lockout duration (minutes): 15 Lockout observation window (minutes): 15 Computer role: WORKSTATION The command completed successfully. C:\Users\SIT> </pre>  I have configured the following: • Set lockout threshold = 5 invalid login attempts • Lockout duration = 15 minutes • Lockout observation window = 15 minutes This prevents brute-force login attempts and enhances resilience against password-guessing attacks. After 5 invalid login attempts, the account will be locked for 15 minutes
6	Disable Internet Connection Sharing	Yong Lin	  <pre> C:\Users\SIT>gpupdate /force Updating policy... Computer Policy update has completed successfully. User Policy update has completed successfully. C:\Users\SIT>sc query sharedaccess SERVICE_NAME: sharedaccess TYPE : 20 WIN32_SHARE_PROCESS STATE : 1 STOPPED WIN32_EXIT_CODE : 1077 (0x435) SERVICE_EXIT_CODE : 0 (0x0) CHECKPOINT : 0x0 WAIT_HINT : 0x0 C:\Users\SIT> </pre> Disabled the Internet Connection Sharing (ICS) service through: • In Service.msc, go to Internet Connection Sharing • Disable Startup Type • Service Status will stop Next, navigate to Computer Configuration → Administrative Templates → Network → Network Connections • Enable “Prohibit use of ICS on your DNS domain network” Lastly, confirm the status via CMD command - sc query sharedaccess

7

Turn on
Protected
File Sharing
and Protect
shared
folders

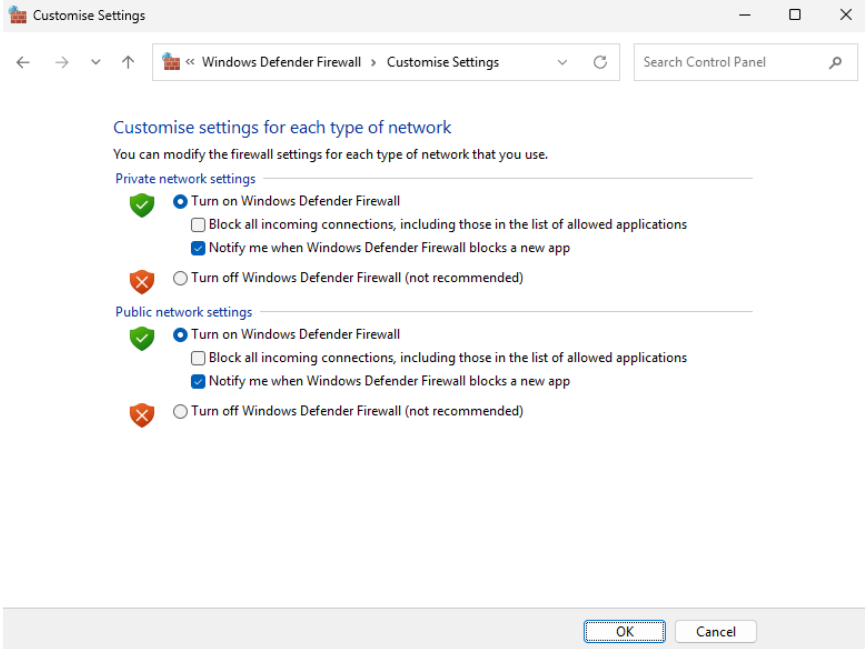
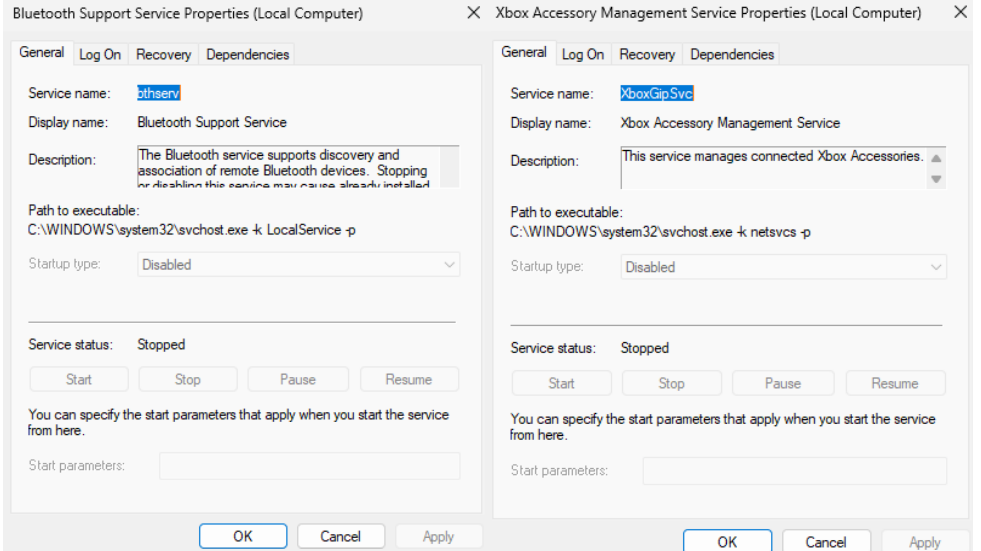
Darren

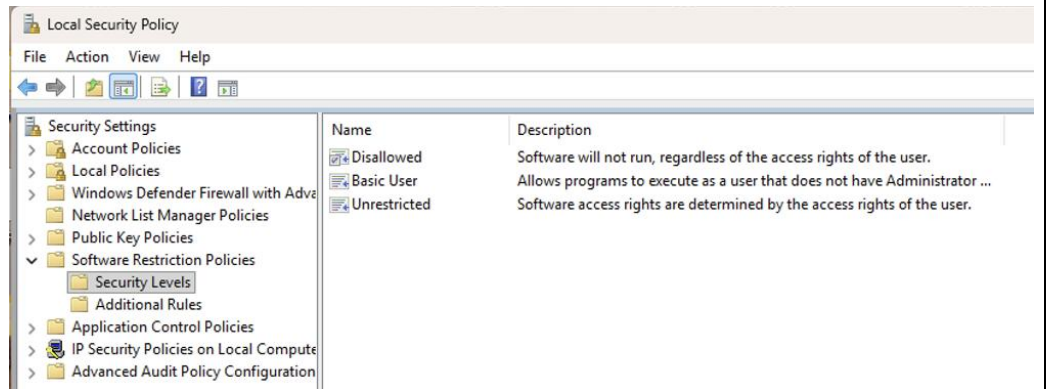


Admin account SIT has full control for the shared folder “share” while Student only has Read access

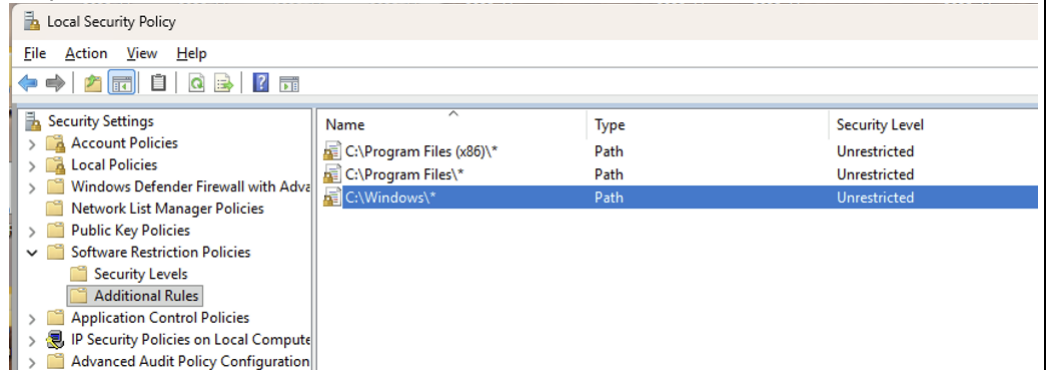
Now, student account has no permissions at all, and cannot access the folder at all

Student account can view the file inside the folder but cannot modify it

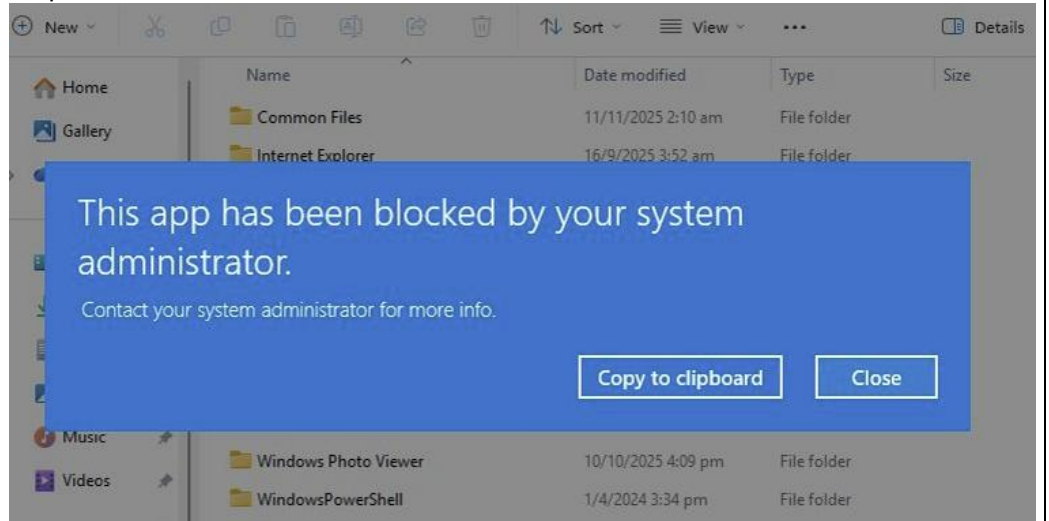
8	Enable Internet Connection Firewall	Darren	
9	Disable unnecessary services	Darren	 Unnecessary services like Bluetooth and Xbox have been disabled
10	Use of software restriction policies where appropriate	Zwe	Step1.



Step2.

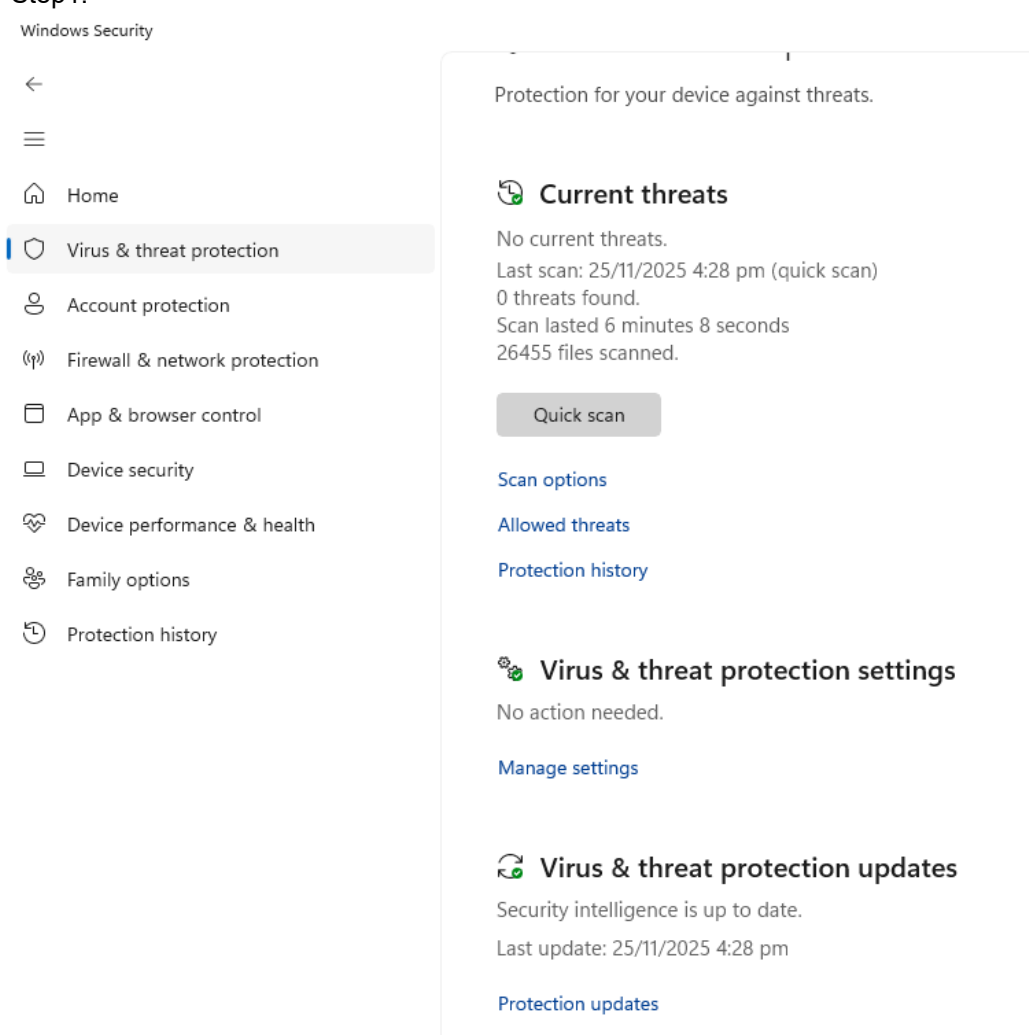


Step3.

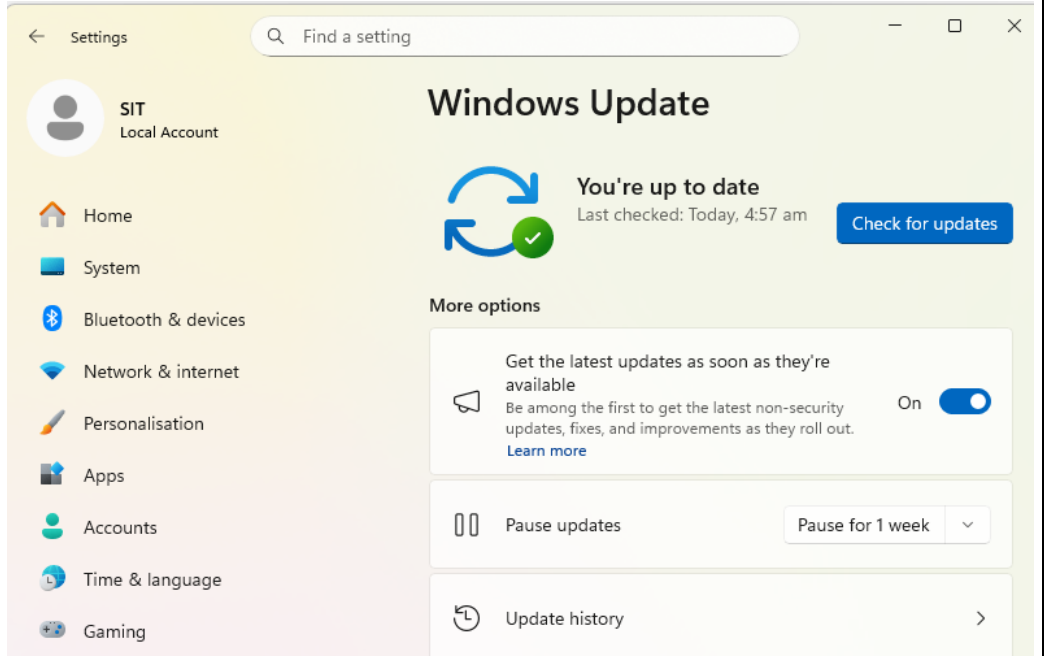


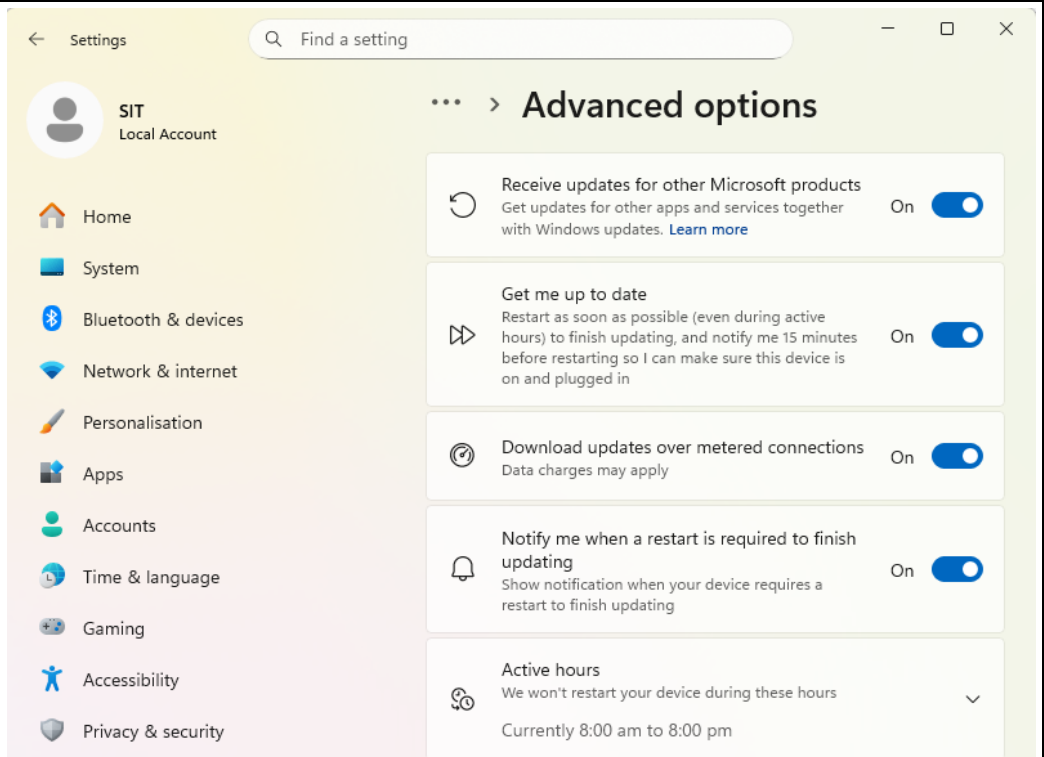
I opened the **Local Security Policy** window by running **secpol.msc** from the Run dialog.

Under **Security Settings**, I went to **Software Restriction Policies** and created a new policy since none existed. Next, I set the default security level to **Disallowed**, which blocks all software unless it is explicitly allowed.

			After that, I added the required Path Rules under Additional Rules to allow essential system folders. For each safe folder (such as C:\Windows*, C:\Program Files*, and C:\Program Files (x86)*), I created a new rule and set its security level to Unrestricted. This ensures that trusted applications inside these locations can still run normally. To apply the changes, I opened Command Prompt as Administrator and ran gpupdate /force. Finally, I tested the policy by attempting to run an .exe from an unapproved folder, and it was correctly blocked. When I moved the same file into <i>C:\Program Files</i>, it ran successfully, confirming that the software restriction policy was working as expected.
11	Install antivirus software and updates	Zwe	Step1.  The screenshot shows the Windows Security application. On the left is a navigation pane with options: Home, Virus & threat protection (selected), Account protection, Firewall & network protection, App & browser control, Device security, Device performance & health, Family options, and Protection history. The main area displays 'Protection for your device against threats.' Under 'Current threats', it states 'No current threats.' and provides scan details: 'Last scan: 25/11/2025 4:28 pm (quick scan)', '0 threats found.', 'Scan lasted 6 minutes 8 seconds', and '26455 files scanned.' There is a 'Quick scan' button. Below are links for 'Scan options', 'Allowed threats', and 'Protection history'. The 'Virus & threat protection settings' section shows 'No action needed.' with a 'Manage settings' link. The 'Virus & threat protection updates' section shows 'Security intelligence is up to date.' and 'Last update: 25/11/2025 4:28 pm' with a 'Protection updates' link. Step2.

		Step3.
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			I opened Windows Security using <i>windowsdefender:</i> from the Run dialog and went to Virus & threat protection. Under <i>Virus & threat protection updates</i>, I clicked Check for updates and waited for the latest security intelligence definitions to finish installing. After that, I checked the Manage settings page and made sure the important toggles were turned on, including Real-time protection, Cloud-delivered protection, and Automatic sample submission, so that Defender can detect threats effectively. To complete the task, I ran a Quick scan to demonstrate the antivirus in action and verified that the scan finished with No current threats.
12	Keep up-to-date on the latest security updates	Zwe	Step1.  The screenshot shows the Windows Settings app, specifically the Windows Update section. The left sidebar lists various settings categories: Home, System, Bluetooth & devices, Network & internet, Personalisation, Apps, Accounts, Time & language, and Gaming. The main content area is titled 'Windows Update' and displays a status message 'You're up to date' with a green checkmark icon. Below this, it says 'Last checked: Today, 4:57 am' and provides a 'Check for updates' button. Under the 'More options' section, there are three main settings: 'Get the latest updates as soon as they're available' (which is turned on), 'Pause updates' (with a 'Pause for 1 week' dropdown), and 'Update history' (with a right arrow). Step2.



Settings

Find a setting

SIT Local Account

Home

System

Bluetooth & devices

Network & internet

Personalisation

Apps

Accounts

Time & language

Gaming

Accessibility

Privacy & security

Advanced options

Receive updates for other Microsoft products
Get updates for other apps and services together with Windows updates. [Learn more](#) On ☒

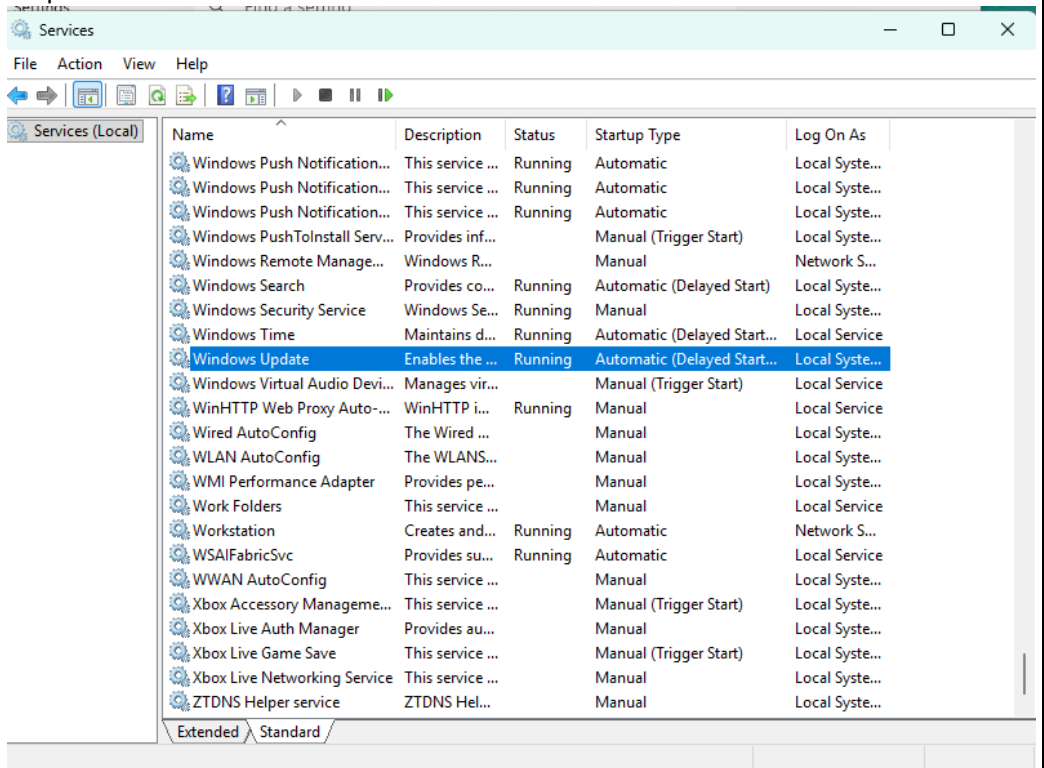
Get me up to date
Restart as soon as possible (even during active hours) to finish updating, and notify me 15 minutes before restarting so I can make sure this device is on and plugged in On ☒

Download updates over metered connections
Data charges may apply On ☒

Notify me when a restart is required to finish updating
Show notification when your device requires a restart to finish updating On ☒

Active hours
We won't restart your device during these hours
Currently 8:00 am to 8:00 pm

Step3.



Services

File Action View Help

Services (Local)

Name	Description	Status	Startup Type	Log On As
Windows Push Notification...	This service ...	Running	Automatic	Local Syste...
Windows Push Notification...	This service ...	Running	Automatic	Local Syste...
Windows Push Notification...	This service ...	Running	Automatic	Local Syste...
Windows PushToInstall Serv...	Provides inf...	Running	Manual (Trigger Start)	Local Syste...
Windows Remote Manage...	Windows R...	Running	Manual	Network S...
Windows Search	Provides co...	Running	Automatic (Delayed Start)	Local Syste...
Windows Security Service	Windows Se...	Running	Manual	Local Syste...
Windows Time	Maintains d...	Running	Automatic (Delayed Start...	Local Service
Windows Update	Enables the ...	Running	Automatic (Delayed Start...	Local Syste...
Windows Virtual Audio Devi...	Manages vir...	Running	Manual (Trigger Start)	Local Service
WinHTTP Web Proxy Auto...	WinHTTP i...	Running	Manual	Local Service
Wired AutoConfig	The Wired ...	Running	Manual	Local Syste...
WLAN AutoConfig	The WLANs...	Running	Manual	Local Syste...
WMI Performance Adapter	Provides pe...	Running	Manual	Local Syste...
Work Folders	This service ...	Running	Manual	Local Service
Workstation	Creates and...	Running	Automatic	Network S...
WSAIFabricSvc	Provides su...	Running	Automatic	Local Service
WWAN AutoConfig	This service ...	Running	Manual	Local Syste...
Xbox Accessory Manageme...	This service ...	Running	Manual (Trigger Start)	Local Syste...
Xbox Live Auth Manager	Provides au...	Running	Manual	Local Syste...
Xbox Live Game Save	This service ...	Running	Manual (Trigger Start)	Local Syste...
Xbox Live Networking Service	This service ...	Running	Manual	Local Syste...
ZTDNS Helper service	ZTDNS Hel...	Running	Manual	Local Syste...

Extended Standard

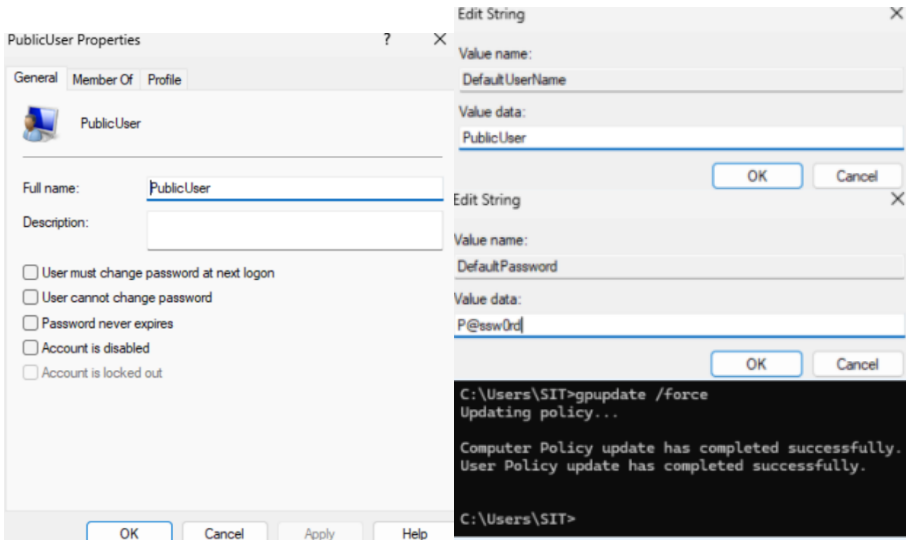
			I opened Windows Update by running <i>ms-settings:windowsupdate</i> from the Run dialog. In the Windows Update page, I clicked Check for updates to manually install all available patches and restarted the system when prompted. After that, I went into Advanced options and made sure all the important toggles were turned on — <i>Receive updates for other Microsoft products</i>, <i>Download updates over metered connections</i>, and <i>Get me up to date</i>, which enables automatic installation of updates. Under Active hours, I set a non-disruptive time window from 8 AM to 8 PM so updates do not interrupt my work. Finally, I opened <i>services.msc</i> and confirmed that the Windows Update service was running with Startup Type: Automatic (Delayed Start) to verify that update services are functioning properly.
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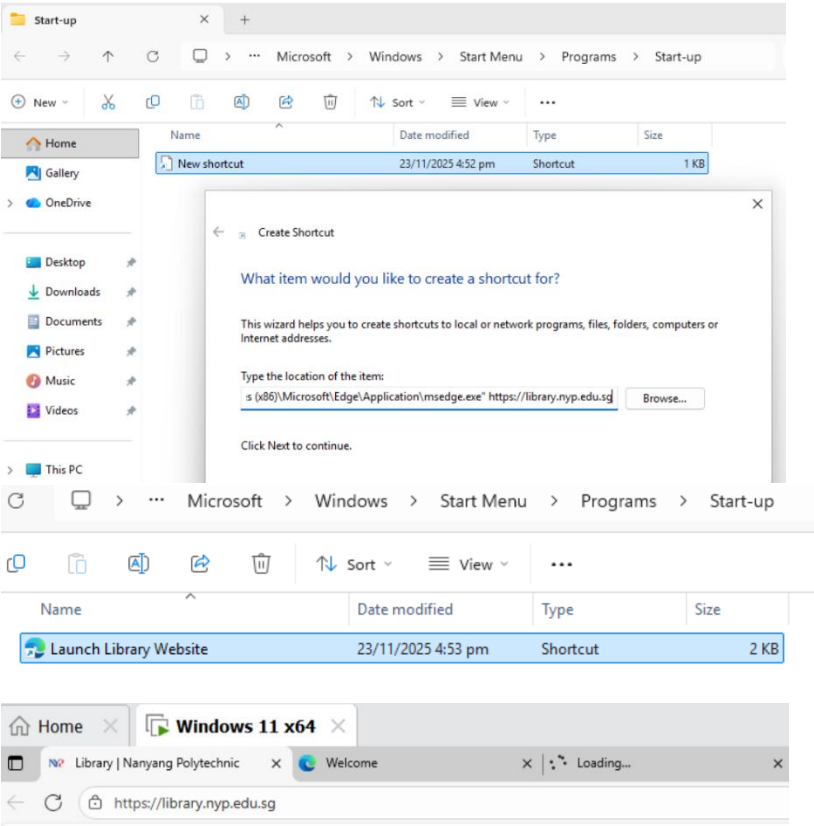
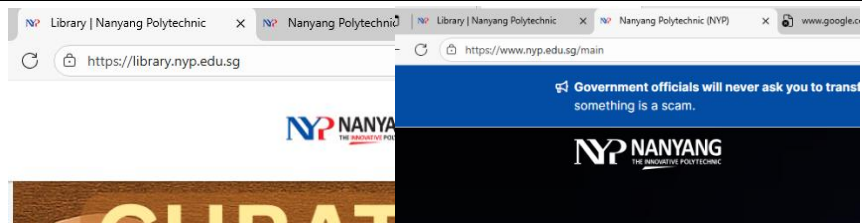
Part 2

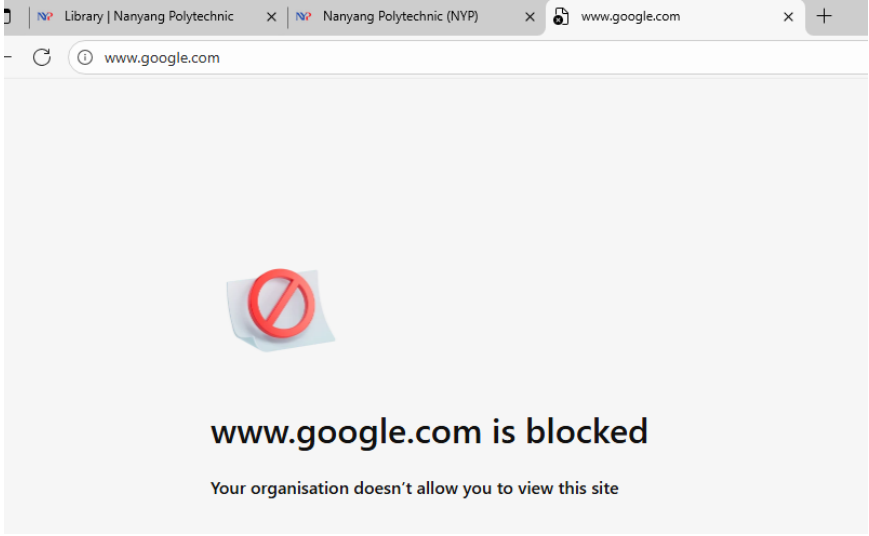
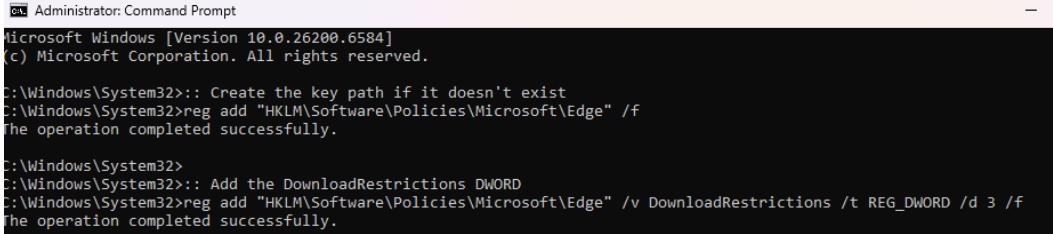
Scenario B – Library Website Terminals

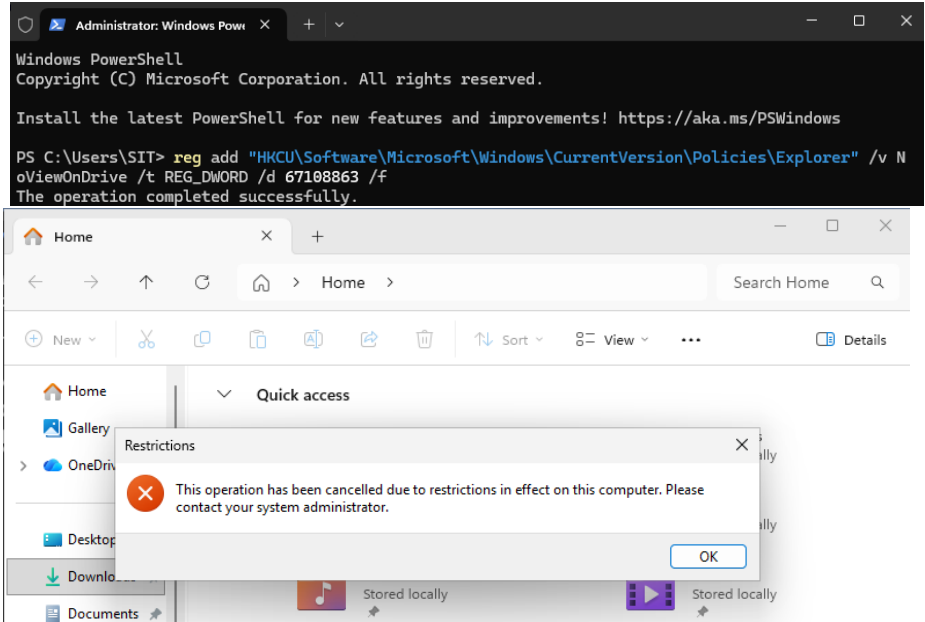
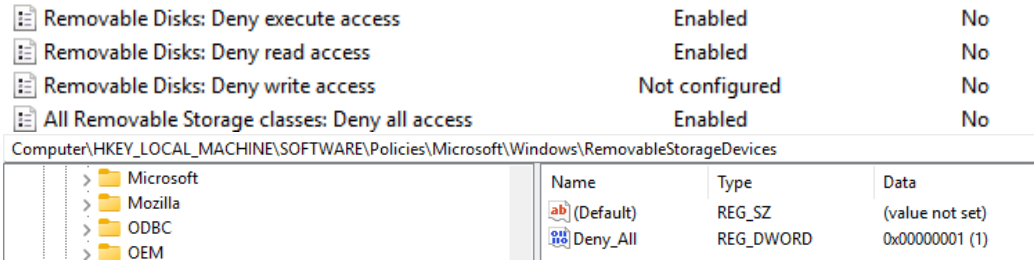
The Polytechnic library plans to implement surfing terminals for students to use the MS Edge browser to access the library website (<https://library.nyp.edu.sg>) and its resources. Students should not have any access to the system except using the MS Edge browser and only the nyp library website and nyp website. Students are not allowed to save any files to local folders, run any other applications, download, install anything, access the USB ports and external drives, shutdown the PC and change the settings of the browser. All unauthorised inbound and outbound connections should be blocked.

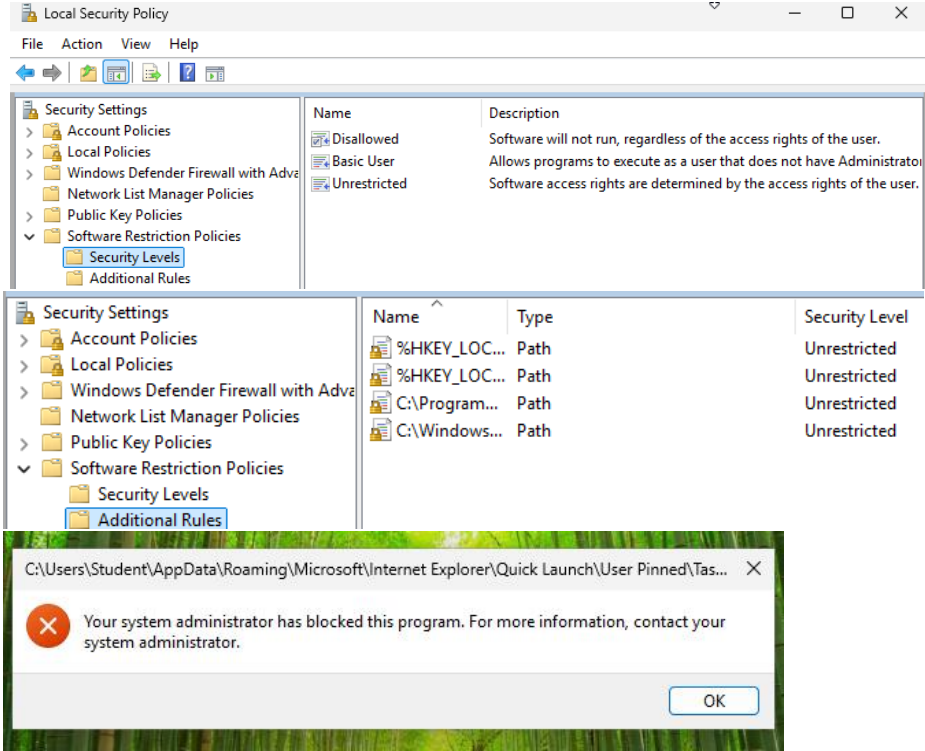
As there are 38 terminals in the library, upon boot-up, it should automatically login to the generic **PublicUser** (use password P@ssw0rd) account and display the nyp library website. It should auto-logout after every 10 mins of inactivity. The terminal should auto shut-down every evening at a fixed time, say 8pm. Write a script such that the administrator can easily run to change to a different auto-shutdown time for the day and put a shortcut to this script on the administrator's desktop screen.

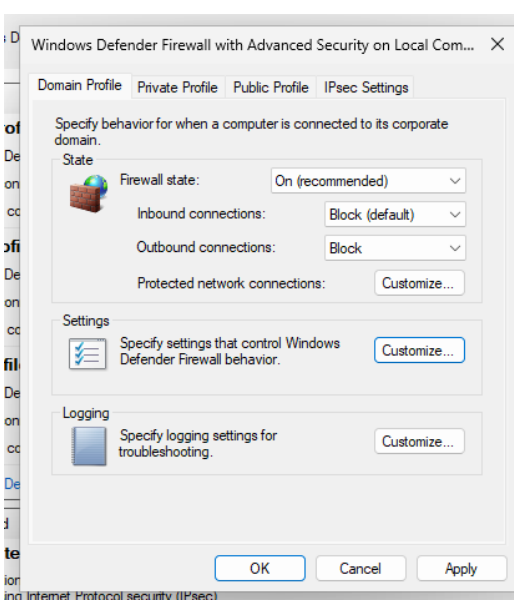
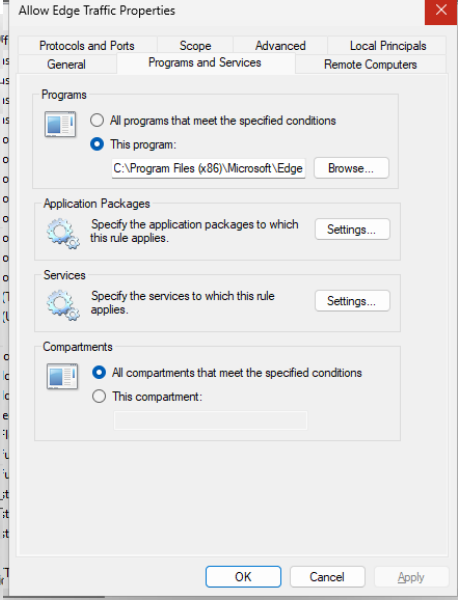
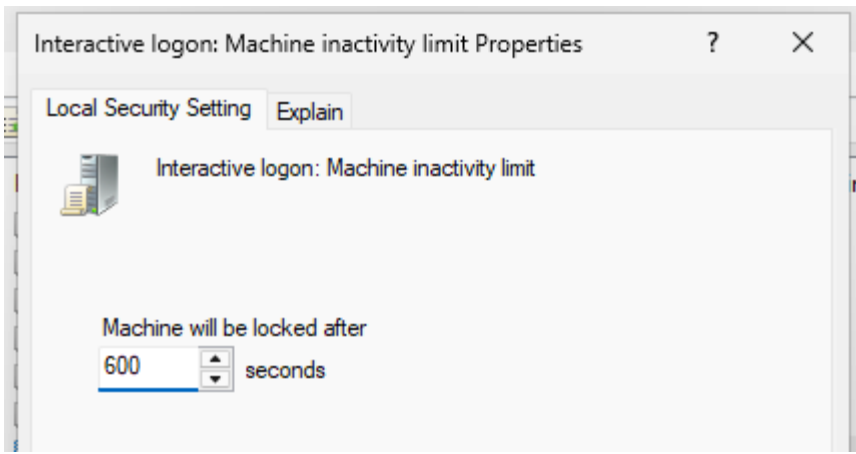
Security Configuration	Done by	Solution implemented (short description and/or screenshot)
Auto-login as PublicUser (use password P@ssw0rd)	Yong Lin	 The screenshot shows the 'PublicUser Properties' dialog box with the 'General' tab selected. The 'Full name' is 'PublicUser' and the 'Description' is empty. The 'User must change password at next logon' checkbox is unchecked. Below this, there are checkboxes for 'User cannot change password', 'Password never expires', 'Account is disabled', and 'Account is locked out', all of which are unchecked. To the right, there are two 'Edit String' dialog boxes. The first one shows 'Value name: DefaultUserName' and 'Value data: PublicUser'. The second one shows 'Value name: DefaultPassword' and 'Value data: P@ssw0rd'. Below these, a command prompt window shows the command 'C:\Users\SIT>gpupdate /force' and the output 'Updating policy... Computer Policy update has completed successfully. User Policy update has completed successfully. C:\Users\SIT>'. At the bottom of the 'PublicUser Properties' dialog, there are buttons for 'OK', 'Cancel', 'Apply', and 'Help'. First, we create a New User - PublicUser and we uncheck “User must change password at next logon” Secondly, enable auto-login via Registry (Window + R → regedit) • AutoAdminLogon → set to 1 • DefaultUserName → set to PublicUser • DefaultPassword → set to P@ssw0rd Next, we run gpupdate /force in CMD to apply the policy Lastly, when we restart the laptop, it will auto-login to PublicUser account with no password entry required

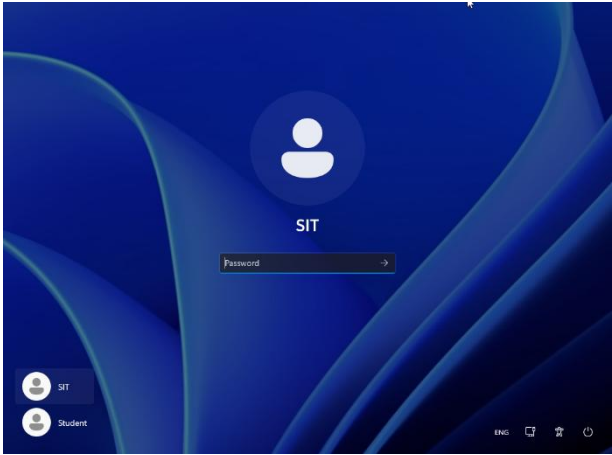
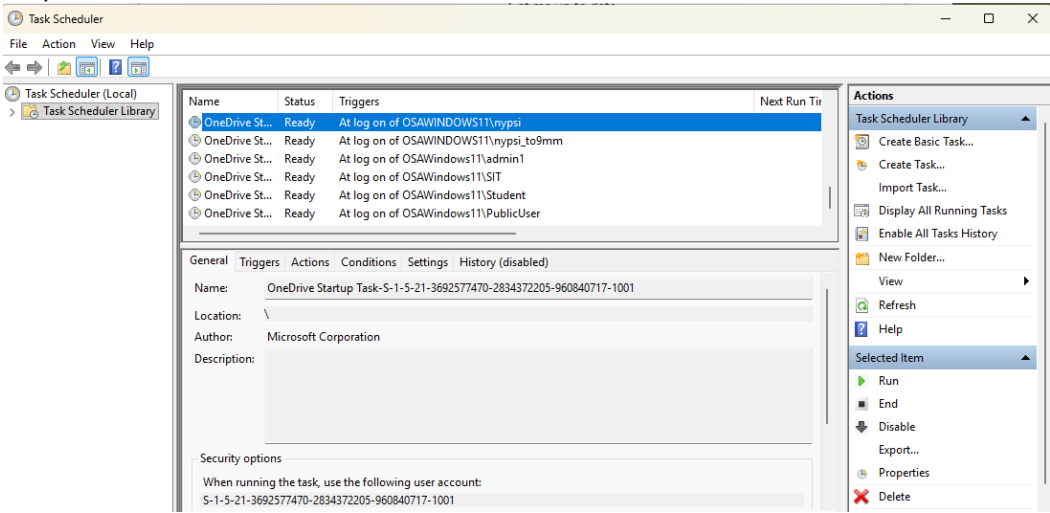
Launch Edge Automatically to NYP library website	Yong Lin	 First, we go to Start-Up in File Explorer and create a shortcut to: "C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" https://library.nyp.edu.sg Next, we just have to restart the laptop and when it auto login PublicUser account, the microsoft edge will auto launch to NYP Library website
Restrict Browser Access to NYP Domains Only	Yong Lin	

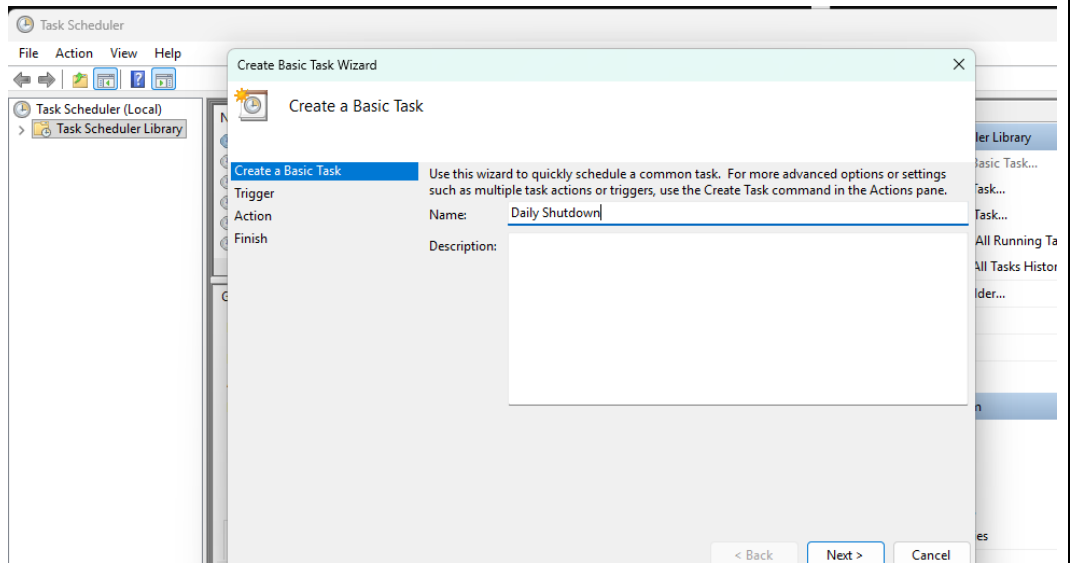
	 First, navigate to User Configuration → Administrative Templates → Microsoft Edge Next, under Defined a list of allowed URLs, click show and under value enter the 2 URLs: • https://library.nyp.edu.sg • https://www.nyp.edu.sg Next, under Block access to a list of URLs, click show and under value enter * (to block all others sites) Finally, we apply the changes by running gpupdate /force in CMD When we use account PublicUser and the edge opens up, we are only able to access the NYP library and NYP websites All other sites like Google will be blocked and show “Your organisation doesn’t allow you to view this site”
Disable downloads and saving files	Jingda  <pre>Administrator: Command Prompt Microsoft Windows [Version 10.0.26200.6584] (c) Microsoft Corporation. All rights reserved. C:\Windows\System32>: Create the key path if it doesn't exist C:\Windows\System32>reg add "HKLM\Software\Policies\Microsoft\Edge" /f The operation completed successfully. C:\Windows\System32> C:\Windows\System32>: Add the DownloadRestrictions DWORD C:\Windows\System32>reg add "HKLM\Software\Policies\Microsoft\Edge" /v DownloadRestrictions /t REG_DWORD /d 3 /f The operation completed successfully.</pre> Couldn't download - Blocked Create the key path if it doesn't exist: reg add "HKLM\Software\Policies\Microsoft\Edge" /f Add the DownloadRestrictions DWORD:

		<pre>reg add "HKLM\Software\Policies\Microsoft\Edge" /v DownloadRestrictions /t REG_DWORD /d 3 /f</pre>
Prevent access to local folders and drives	Jingda	 Restrict view on drive for file explorers <pre>reg add "HKCU\Software\Microsoft\Windows\CurrentVersion\Policies\Explorer" /v NoViewOnDrive /t REG_DWORD /d 67108863 /f</pre>
Block USB Ports and external drives	Jingda	 Go to gpedit.msc Local Computer Policy > Administrative Templates > System > Removable Storage Access Enable: - Removable Disks: Deny Read Access - Removable Disks: Deny Write Access - All Removable Storage Classes: Deny All Access To verify: Registry Editor HKEY_LOCAL_MACHINE\SOFTWARE\Policies\Microsoft\Windows\RemovableStorageDevices

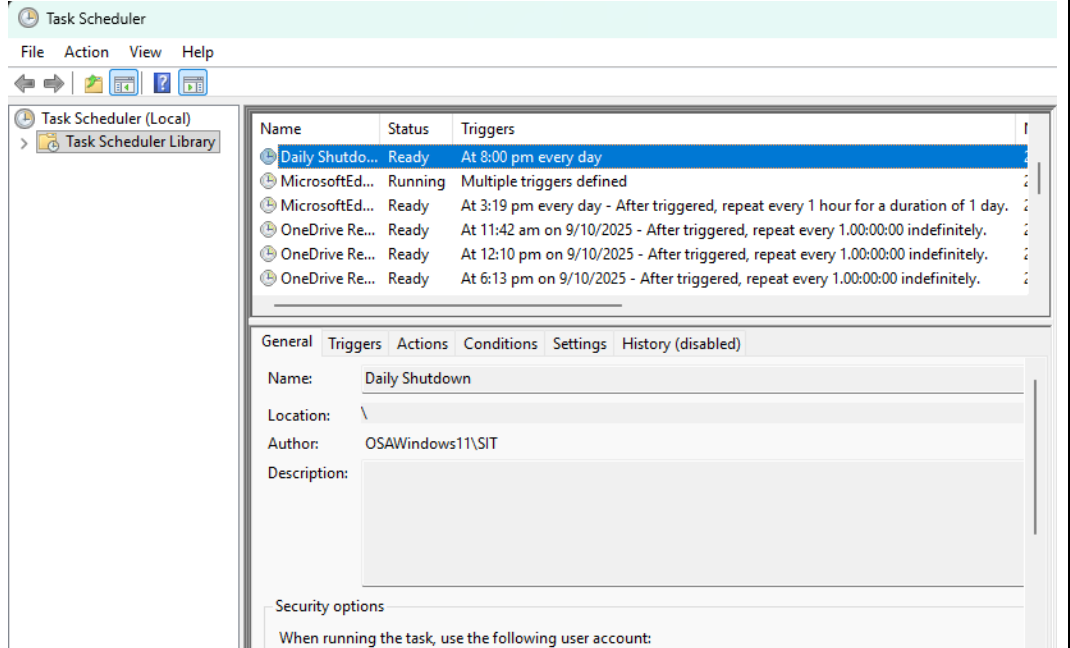
		Result: USB storage devices won't be usable at all Students cannot download or save files into USB storage devices No external installable software can run in/from USB storage devices
Restrict running other applications	Darren	 Default security level has been set to Disallowed, meaning no applications should be able to run outside the admin account Exception given to Windows programs, MS Edge and CMD to allow testing Any programs outside the path exceptions earlier, will not be able to run, whereas programs included in the exceptions will be able to run

Block unauthorised inbound and outbound connections	Darren	  <pre>C:\Windows\System32>ping google.com Pinging google.com [172.253.118.102] with 32 bytes of data: General failure. General failure. General failure. General failure. Ping statistics for 172.253.118.102: Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),</pre>
Auto-logout after 10 minutes of inactivity	Darren	

		 The image shows a Windows 10 login screen. The background is a blue abstract design. In the center, there is a white user icon and the text 'SIT' below it. Below the name is a password field with a 'Password' label and a right-pointing arrow. In the bottom-left corner, there are two user icons: 'SIT' and 'Student'. In the bottom-right corner, there are icons for language (ENG), network, volume, and power.
		Set auto-log to 10 minutes after idling, the system logs out
Auto-shutdown daily at 8pm	Zwe	Step1.  The image shows the Windows Task Scheduler application. The 'Task Scheduler Library' is expanded on the left. The main pane shows a list of tasks. The task 'OneDrive Startup Task-S-1-5-21-3692577470-2834372205-960840717-1001' is selected. The 'General' tab is active, showing the task name, location, author, and description. The 'Security options' section shows the user account 'S-1-5-21-3692577470-2834372205-960840717-1001'. Step2.



Step3.



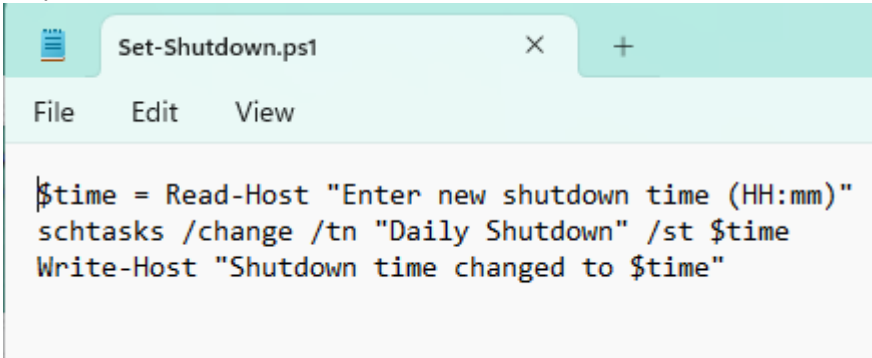
I opened **Task Scheduler** from the Start menu and created a new **Basic Task** named *Daily Shutdown*. For the trigger, I set it to run **daily at 8:00 PM**. In the Action step, I selected **Start a Program** and entered shutdown as the program/script. Under arguments, I added `/s /f /t 0` so the system will shut down immediately and force close any apps.

After finishing the setup, I checked the **Task Scheduler Library** to verify that the task was created successfully and scheduled to run every day at 8 PM.

Admin script to
change
shutdown time

Zwe

Step1.

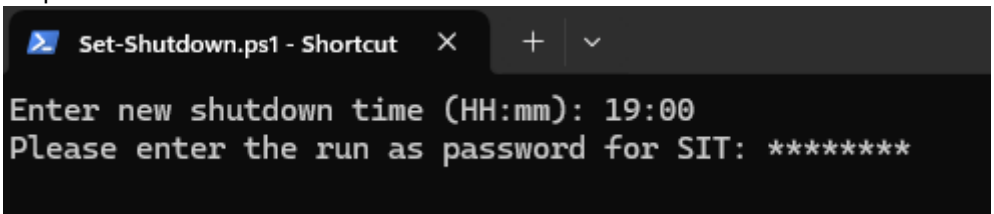


```

$time = Read-Host "Enter new shutdown time (HH:mm)"
schtasks /change /tn "Daily Shutdown" /st $time
Write-Host "Shutdown time changed to $time"

```

Step2.

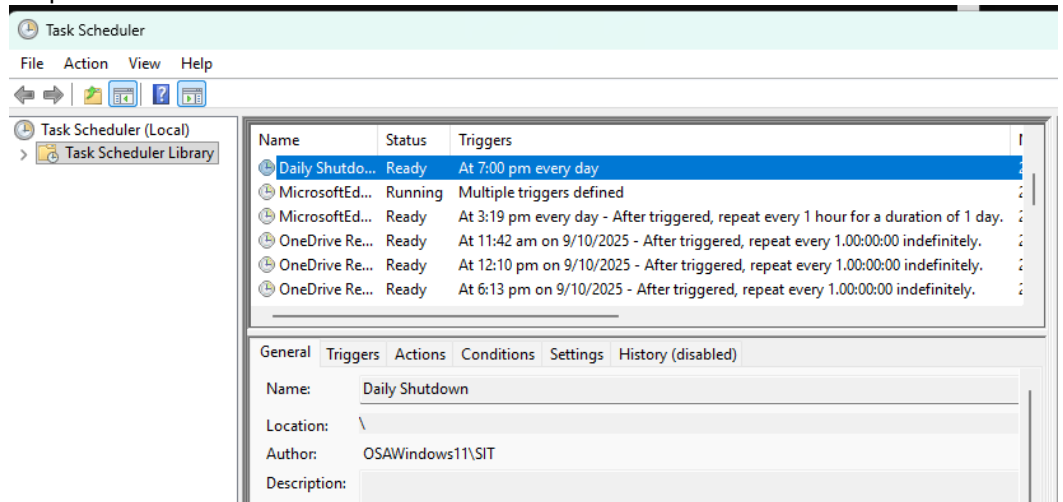


```

Enter new shutdown time (HH:mm): 19:00
Please enter the run as password for SIT: *****

```

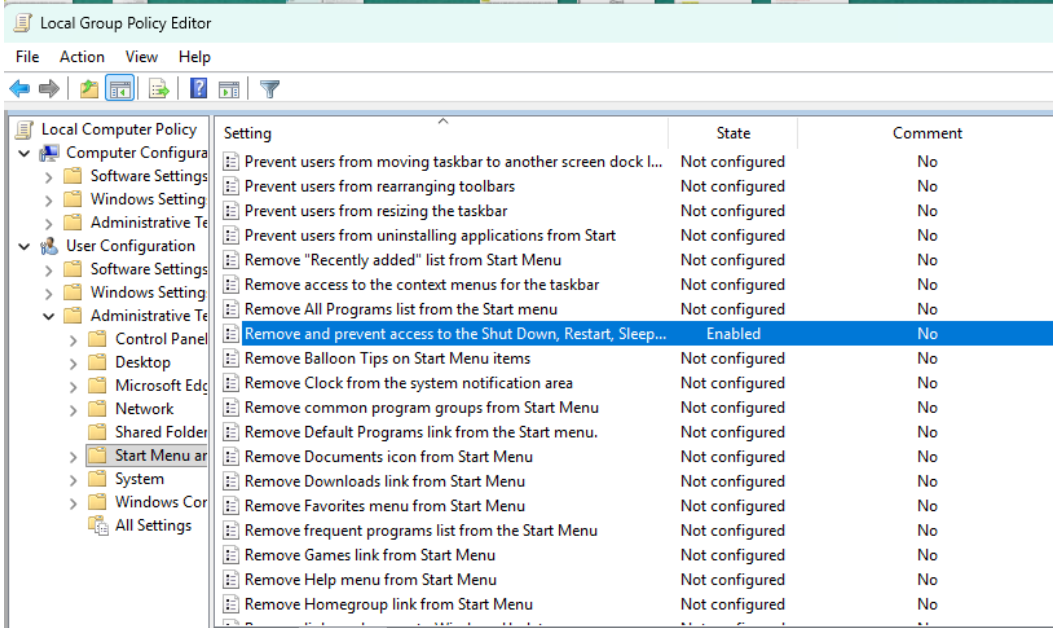
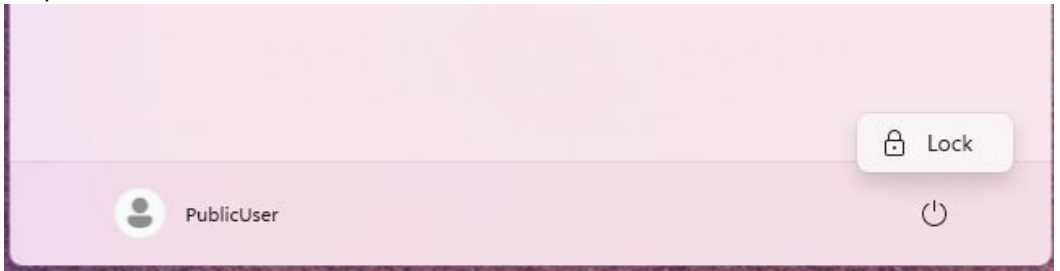
Step3.



I created a PowerShell script that allows the shutdown time of the daily task to be updated easily.

First, I opened **Notepad** and pasted the PowerShell code that prompts the user for a new time and applies it using the `schtasks /change` command. I saved the file on my Desktop as **Set-Shutdown.ps1**.

Next, I created a **shortcut** to the script so it can be run with administrative privileges. In the shortcut's *Target* field, I entered:

		powershell.exe -ExecutionPolicy Bypass -File "%USERPROFILE%\Desktop\Set-Shutdown.ps1" This ensures the script runs even if PowerShell's execution policy is restricted. After setting the shortcut to Run as administrator, I tested it by running the shortcut and entering a new shutdown time as 19:00 from the originally set 20:00. The script updated the Task Scheduler entry successfully and confirmed the new time.
Prevent shutdown from PublicUser Account	Zwe	Step1.  Step2.  I opened the Local Group Policy Editor by running gpedit.msc. Under <i>User Configuration</i> → <i>Administrative Templates</i> → <i>Start Menu and Taskbar</i>, I looked for the policy named <i>Remove and prevent access to the Shut Down, Restart, Sleep, and Hibernate commands</i>. I enabled that policy, so that the PublicUser account will no longer see or use any shutdown-related options. After applying the changes, I ran gpupdate /force to refresh the policy. This ensures the restrictions take effect immediately for the PublicUser account.

